Leveraging Machine and Deep Learning Technology for Prediction of Bridge Deck Condition

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Objective

The attributes of a structural bridge such as the Superstructure, Substructure, Deck, and Culvert are considered as key elements that determine the overall structural and functional health of any bridge. To develop and implement effective maintenance schedules for the various road networks, deterioration models for these critical bridge elements need to be established. Factors such as creep, shrinkage, cracking, fatigue, reinforcement corrosion, environment, average daily traffic, operational practices, etc., interact to collectively affect the deterioration rates and future condition of bridge elements. Using mechanistic models to predict the service life of structural elements is a huge challenge due to the complexity in quantifying the contributions of these underlining factors. Stochastic models and artificial intelligence models are the most practical ways to predict the condition of bridge elements using previously collected data. Artificial Intelligence methods such as Neural Networks can be an intuitive method to determine the condition of key elements of bridge by making use of suitable algorithms for predicting these above variables. In some cases, a linear model may suffice, but when compared to Artificial Neural Network the accuracy is slightly inferior. (Ali, Gasser, et al, 2019)

This study aims to determine the best model to predict the condition of the bridge deck. In pursuit of our objective, several algorithms shall be tested using a base-

dense neural network to determine which algorithm predicts the condition of the bridge with highest accuracy. A custom designed dense neural network shall be developed using hyperparameter tuning, manipulating the number of neurons and layers, and early stopping to increase the accuracy. An assumption has been made that the strength of the algorithm's correlation with predicting deck condition can be determined using the base-dense neural network. The algorithms that will be used for determining which factors influence the condition of the deck are (i) Random Forest, (ii)Generalised Linear Method, (iii) Ordinal Logistic Regression, (iv)Ordinary Least Squares, (v)Principal Component Analysis, and (vi)Negative Binomial. These will be implemented using in-built libraries of Scikit Learn and Statsmodels packages in Python. The prediction was done using Dense Neural Networks (DNN) with a similar architecture for all the algorithms after determining the top 25 and top 5 influencing parameters. The algorithm which gave a higher accuracy was chosen and further architecture manipulation and hyperparameter tuning was performed.

Methodology

The flowchart shown in Figure 1 briefly describes the research and modeling approach applied in this study.

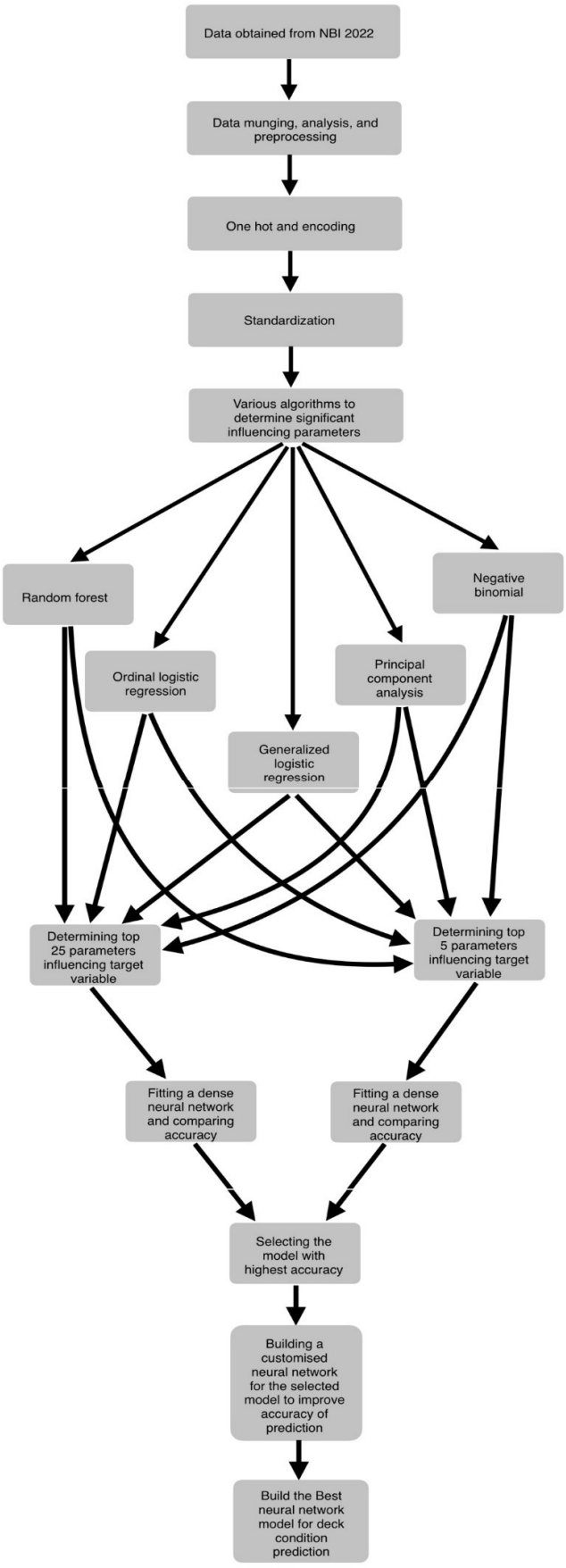


Figure 1: Flowchart demonstrating the research and modelling methodology

Data Description

The data for analysis in this paper was obtained from the U.S Department of Transportation, Federal Highway Administration, 2022. Federal legislation and publications by the U.S. Department of Transportation, Federal Highway Administration (FHWA) require that bridge data herein specified be collected and maintained for all bridges on the Public Highway and Street System in Ohio. Accomplishing this task will be possible only through the complete cooperation of all government and local agencies involved.[1]

Data Munging, Analysis and Preprocessing

Data obtained from Federal Highway Administration has been analyzed and necessary data manipulation has been done to develop a model with optimum accuracy for prediction of deck condition. Python packages such as Pandas, Matplotlib, Seaborn, Sklearn, Statsmodels were used for data manipulation, analysis, and training of the dataset. The dataset initially consisted of 27003 bridges recorded in rows of the datasheet and 123 features recorded in the columns. After data preprocessing, which included preparation techniques such as removal of features where values are null, too few values recorded, and blank items, the dataset was reduced to 27003 bridges with 108 features.

The columns that were dropped for various reasons

Feature Name	Reason for dropping
WORK_PROPOSED_075	Too many null values
WORK_DONE_075B	Too many null values

[1] (Ohio Bridge Inventory Coding Guide,2021, https://www.transportation.ohio.gov/working/engineering/structural/inspection/03-inventory-coding).[1]

LRS_INV_ROUTE_013A	Too many null values
SUBROUTE_NO_013B	Too many null values
SPEC_LAST_DATE_093C	Too many null values
UNDWATER_LAST_DATE_093B	Too many null values
FRACTURE_LAST_DATE_093A	Too many null values
YEAR_OF_IMP_097	Too many null values
OTHER_STATE_CODE_098A	Too many null values
OTHR_STATE_STRUC_NO_099	If items 98A and 98B are blank then this value must be blank
CRITICAL_FACILITY_006B	Too many null values
TEMP_STRUCTURE_103	Too many null values
MIN_NAV_CLR_MT_116	Too many null values
WORK_PROPOSED_075A	Too many null values
OTHER_STATE_PCNT_098	Too many null values

Table 2: Imputations performed on features that had few null values from 444 in PERCENT_ADT_TRUCK_109 to 3 in BRIDGE_IMP_COST_094 are tabulated below,

Feature Name	Action Taken
PERCENT_ADT_TRUCK_109	Replaced 444 null values with 0
TOTAL_IMP_COST	Replaced 98 null values with 0
ROADWAY_IMP_COST_095	Replaced 91 null values with 0
IMP_LEN_MT_076	Replaced 17 null values with 0

YEAR_RECONSTRUCTED_106	Replaced 13 null values with 0
BASE_HWY_NETWORK_012	Replaced the 5 null values with 0
PIER_PROTECTION_111	Replaced the 4 null values with 1 – navigation control not needed
BRIDGE_IMP_COST_094	Replaced the 3 null values with 0

The data is randomly split into training and test sets, forming 80% and 20%, respectively. Data Preprocessing is performed to normalize it. The pre-processing normalization method allows the data to be between zero and one.

$$f(x_k) = \frac{x_k - min(x_k)}{max(x_k) - min(x_K)} \quad \text{(Equation 1)}$$

(Ali, Gasser, et al 2019)

One Hot Encoding and Label Encoding

In our datasheet, some variables are categorical such as STRUCTURE_KIND_043A and other columns are numeric such as PERCENT_ADT_TRUCK_109. Hence, the categorical data was converted to numeric data using Label Encoder.(Vincent, Durai Raj, et al,2019). Label encoding is preferred over one–hot encoding since, one–hot

encoding considerably increases the dimensionality of the data and label encoding provides fewer errors. (Chugh, Gunjan, Shailender Kumar, and Nanhay Singh,2021).

Standardization and Normalization

The input data needs to be standardized to avoid wider ranges of data from dominating the distance metric and facilitate distance learning metric (Srinivasan, Dipti, A. C. Liew, and C. S. Chang). The process of converting categorical data and normalizing numerical data is called Standardization. Data Standardization ensures that distributions of all variables are zero–centered and hence avoids skewed observations (Assaad, Rayan, and Islam H. El-Adaway,2020). Neural Network's training time and accuracy are impacted by the process of standardization. The input layer consists of a number of neurons equal to the input features that are being used for prediction. (Althaqafi, Essam, and Eddie Chou,2022). In this study standardization of input data was performed using the Z-score, which is expressed as follows,

$$z = \frac{\bar{x} - \mu}{\sigma} \quad (Equation\ 2)$$

Where $\bar{x}$ is non – standardized value, μ is the mean value, and σ is the standard deviation.

After data preprocessing, Standardized normalization, and label encoding, 27003 bridges were used to train the neural network using various algorithms.

Feature Selection

A subset of features shall be used in a way that includes the variables and features that contribute the most to the predicted variable. Feature Selection is a process of selecting the most useful features for building models with better generalization ability(Solorio-Fernández, Saúl, J. Ariel Carrasco-Ochoa, and José Fco Martínez-Trinidad,2020)

Selecting features that affect the target variable facilitates the smooth functioning of Artificial Neural Network model. Essential features that affect the bridge deck condition were selected and used as input features to the artificial neural network. (Ayyub, Bilal M., and Richard H. McCuen,2003). The top 25 and 5 features that affect the bridge condition using several algorithms are selected to ascertain the accuracy of prediction using Artificial Neural Network. The parameters that have a 95% confidence level of influence or more are utilized as input features for ANN. This can be judged using the p – value from the statistical table obtained after regressing fitting for respective algorithms. If the p – value is less than 0.05 than that parameter is said to be influencing the target parameter with a 95% confidence interval.

Artificial Neural Network (ANN)

Artificial Neural Networks are some of the most widely used artificial intelligence algorithms that recognize the pattern between input and output variables.(Dawood, Thikra, Zhenhua Zhu, and Tarek Zayed,2018). Artificial Neural Networks uses different mathematical layers to learn various features in the data that is being processed. An ANN can contain multiple layers with thousands of neurons arranged in layers called units. The input layer receives multiple types of

data and extracts various features of data, the extracted information is then passed on to the hidden layers. The hidden layer is where the learning and transformation process takes place. The information is then passed on to the output layer where the original format is now retained.(Althaqafi, Essam, and Eddie Chou,2022). ANNs learn by experience through error minimization by adjusting the weights and biases. The Adaptive moment estimation solver has been used since evidence shows that they are better than classical stochastic gradient descent.(Assaad, Rayan, and Islam H. El-Adaway,2020). The configuration of neural network highly affects their performance,(Negnevitsky, Michael,2005) and hence the keras tuner has been used over the base – neural network to improve the performance of the neural network. The architecture for a neural network generally consists of (1) input layer that accepts input signals and redistributes them to all neurons in the hidden layers, (2) one or more hidden layers which include computing neurons for the processing of input patterns and (3) an output layer that accepts output signals from hidden layers and reports output pattern of entire architecture(Negnevitsky, Michael,2005). The number of neurons in the input layer will equal the input features and the number of neurons in the output layer will equal the number of outputs.(Assaad, Rayan, and Islam H. El-Adaway,2020). In this paper, the output is a prediction of the condition of the bridge from 2 to 10 ('N' refers to not applicable, has been replaced in the datasheet with 10).

Multi-Layer Perceptron

A multi-layer perceptron consists of individual or multiple layers of neurons, which is a feed – forward type of neural network. A multi-layer feed forward neural network that is trained through network back propagation where a network is established between the input and the output depending on the input features. (Mahmoud, Omer, Farhat Anwar, and Momoh-Jimoh E. Salami, 2007).

Activation Function

When the sigmoid hidden units in the feed forward mechanism are replaced with piecewise linear hidden units such as Rectified Linear Units (ReLU) which are most common activation functions used in Neural Networks, the functionality of feed-forward networks has improved.

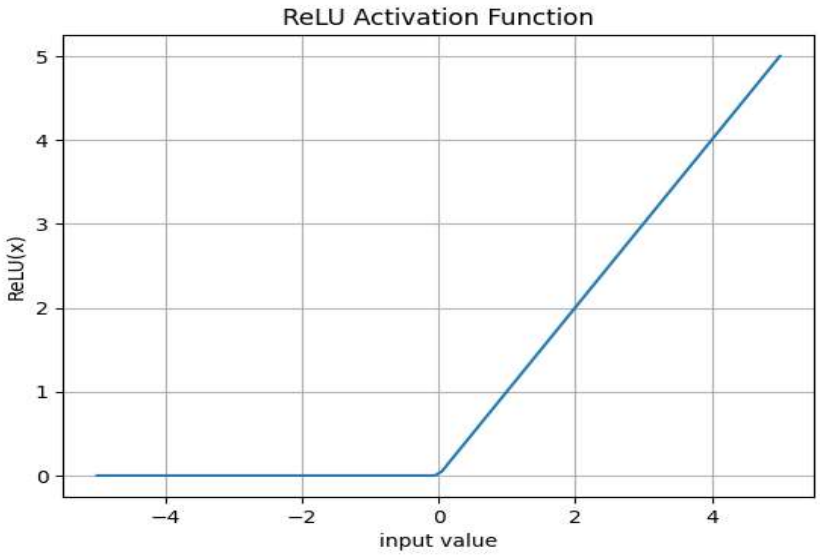


Figure 2 : RELU Activation Function

$$\sigma(z) = max(0, z) = \{ \quad 0 \; if \; z \; < 0 \; and \; z \; if \; z \geq 0 \quad \}$$ (Equation 2)

Here, when z < 0 the function is zero and when z is greater than or equal to zero the function is z. This function although considered. (Althaqafi, Essam, and Eddie Chou,2022).

A base-dense neural network that is used for all the algorithms for the prediction of the condition of bridge, the initial architecture has been defined as below to choose the algorithm that performs with a higher accuracy based on the base–network architecture.

After the algorithm that performs with a higher accuracy than the other algorithms has been determined, other combinations to modify the existing algorithm has been used.

Table 3: Hyperparameter Tuning for improving Accuracy

Hyperparameters	Combination_1	Combination_2_Keras_Tuner	Combination_3_Keras_Tuner
Use Cases	Random Forest Ordinal Logistic Regression Ordinal Least Squares	Negative Binomial	Negative Binomial
Input Neurons	23	23	23
Intermediate layers	3	3	3
Intermediate Neurons Layer 1	46	46	46
Intermediate Neurons Layer 2	46	46	46
Intermediate Neurons Layer 3	23	23	23
Output Layer Neurons	9	9	9
Loss Function	Sparse Categorical Entropy	Sparse Categorical Entropy	Sparse Categorical Entropy

Input Activation Function	Relu	Relu	Relu
Intermediate Activation Function	Relu	Relu	Relu
Output Optimization Function	Adam	Adam	Adam
Number of Epochs	100	100	100
Batch Size			
Validation Split	0.2	0.2	0.2
Accuracy	Ranges between 37 to 80 %	73%	80%

Table 4: Summary of some of the important model variables

Variable	Description	Role
DECK_COND_058		Output
STATE_CODE_001	Three digit code where first two digits are Federal Information	Input

	Processing Standards (FIPS) and third digit is Federal Highway Authority region code.	
RECORD_TYPE_005A	Code to indicate if the structure carries the route or goes under it.	Input
ROUTE_PREFIX_005B	Code for inventory route from 1 to 8.	Input
SERVICE_LEVEL_005C	Code to identify designated level of service.	Input
ROUTE_NUMBER_005D		Input

DIRECTION_005E	Code for the directional suffix to the route number when (1) is part of the route number.	Input
HIGHWAY_DISTRICT_002	Code to indicate the highway agency district in which the bridge is located.	Input
COUNTY_CODE_003	Counties shall be identified based on the Federal Information Processing Standards	Input

PLACE_CODE_004	Census designated places can be identified using the FIPS.	Input
FEATURES_DESC_006A	Name or names of inventory route intersected by inventory route at the structure.	Input
FACILITY_CARRIED_007	The facility being carried shall be recorded and coded.	Input
LOCATION_009	Description of bridge location is recorded here.	Input

MIN_VERT_CLR_010	Practical maximum clearance is recorded here.	Input
KILOPOINT_011	County Log Point (CLP) in miles at start of bridge.	Input
BASE_HWY_NETWORK_012	Coding is done based on whether the inventory route is on Base – Network or not.	Input
LAT_016	Latitude of the bridge location	Input
LONG_017	Longitude of bridge location	Input
DETOUR_KILOS_019	Additional travel	Input

	length because of closure of bridge.	
TOLL_020	Coding shall be done based on whether toll on the route exists.	Input
MAINTENANCE_021	Coding based on the agency responsible for maintenanc e of bridge	Input
OWNER_022	The agency that is the primary owner of the bridge	Input
FUNCTIONAL_CLASS_026	Functional classificatio n of inventory route	Input

YEAR_BUILT_027	The year when the bridge was first built will be shown here	Input
TRAFFIC_LANES_ON_028A	Code the number of lanes of traffic being carried by the structure.	Input
TRAFFIC_LANES_UND_028B	Code the number of lanes of traffic being crossed over by the structure.	Input

Table 4: Federal Highway Administration ratings for bridge conditions (FHWA 1995)

Code	**Description**
N	Not Applicable
9	Excellent Condition
8	Very Good Condition

7	Good Condition
6	Satisfactory Condition
5	Fair Condition
4	Poor Condition
3	Serious Condition
2	Critical Condition
1	Imminent Failure Condition
0	Failed Condition

Literature Review

Deterministic and Stochastic methods are the two major methods that are popularly used for understanding the behaviour of a bridge with time. Deterministic deterioration models are the best models for describing bridge deterioration and incorporate regression model equations associated with explanatory variables. (Veshosky, David, et al,1994). Markov chain is the most generalized model that is used to comment on the condition deterioration of a bridge. (Agrawal, Anil K., Akira Kawaguchi, and Zheng Chen,2010).

Data for our project has been obtained from the National Bridge Inventory of Ohio State.

Sometimes the data for all the bridges are unavailable; hence, the data of the available bridges are clustered, and the data deficient bridges information is extrapolated. The Gaussian Process Regression is used to cluster data with similar deterioration trends. (Dhada, Maharshi, Georgios M. Hadjidemetriou, and Ajith K. Parlikad, 2020). The software has also been developed so that the conditions of various bridges can be showcased at any point in time. BRIDGIT bridge management

software system is one such instance that uses the Markov chain model to identify a bridge element's condition state.

Bridges consist of different elements whose condition health is essential to maintain the bridge performance. There are more than 25 components of the bridge for which the data needs to record by the bridge inspector. These are the primary components of the bridge. The members of the bridges deteriorate with time depending on various conditions such make of the bridge - Steel, R.C.C. or P.S.C., and epoxy-coated bars (Agrawal, Anil K., Akira Kawaguchi, and Zheng Chen, 2010). The deterioration of bridges can also be because of poor material properties (shrinkage), inadequate pouring procedures, freeze and thaw cycles, truckload spectra (axle loads and groups of axles), fatigue and degradation of concrete under compressive load cycles, corrosion of rebars and usage of dicing chemicals (Chang, Minwoo, Marc Maguire, and Yan Sun,2019)). Cracks in a bridge can develop transverse, longitudinal, and diagonal. The Freeze and Thaw cycles cause map cracks oriented in different directions. Locally weaker concrete allows water to penetrate the steel rebars and freeze in low temperatures.

There are different ways of testing on the site through which the condition of the structural elements of the bridges can be ascertained, one of them being the non-destructive tests. Some of the most common non-destructive tests are the Impact Echo, Ground Penetration Radar, Half Cell Potential, Ultrasonic Waves Method, and Electrical Resistivity Method. The deterioration curves can be established for these methods for the condition index and Age of the bridge. For example, the study of the deck of one particular 'Haymarket Bridge' using the above-mentioned non-destructive testing in New York City has revealed that the bridge has excessively

damaged in 6 years at a disproportionately higher rate than its previous deterioration. (Gucunski, Nenad, et al,2017)

Corrosion of the bridge components is one of the dictating parameters in the condition of a bridge. The bridges are rated in Ohio based on the structural and functional health of the bridge from 1 to 9. A rating of 7 means that the bridge is in an ideal condition and needs no intervention toward maintenance then. A bridge rating of 2 to 3 is the worst condition, requiring immediate repair, rehabilitation, or retrofitting. A rating of 1 would mean in a failed state, and a rating of 8 or 9 is assigned to bridges that are 'not applicable' and 'condition and existence unknown,' respectively. A biennial audit is done on the structural and functional health of the bridge, and the appointed bridge inspector awards a condition rating. Some bridges that are in a more debilitating condition are assessed every year. Department of Transportation uses bridge condition ratings to classify structural deficiencies and functional obsolescence. (Marchione et al. 2017).

The *Corrosion Anomaly rate* of the bridges is a parameter that is assessed to judge the bridge's condition. Section losses and cracks occur in the steel girders of an R.C.C. deck bridge, and the state of these components can be assessed using the Corrosion Anomaly Rate (C.A.R.). The Corrosion Anomaly Rate is defined as the number of corrosion anomalies per girder length and observation years. The total distance of girders equals the length of the girder times the number of spans. (Prasad, Avinash, Ilan Juran, and Bojidar Yanev,2022). The coating of the reinforcement used in the construction of the bridge can have an impact on the deterioration trend of the bridge. The deterioration plots for the uncoated rebars are steeper than the coated bars using the Markov and Weibull regression approaches.(Agrawal, Anil K., Akira Kawaguchi, and Zheng Chen, 2010).

Various analyses of the bridge data have been performed to determine how the condition ratings of a bridge get affected by the different parameters of a bridge. To calculate the bridge deterioration, a comparison has been made by Weibull and Markov chain process to know a bridge's decline rate. The Weibull-based approach performs better regarding the experimental conditions than the traditionally used Markov chain for developing deterioration curves for different bridge elements. (Agrawal, Anil K., Akira Kawaguchi, and Zheng Chen,2010). The reasons for bridge deterioration are reinforcement corrosion, concrete degradation, creep, shrinkage, cracking, fatigue, etc. To make a comparison among different parameters as to which would perform better depends on the study period. The deterioration model developed by optimization, percentage prediction, and logistic regression performs better than the other two when the period is more significant than ten years. The optimization technique performs better when the study time is less than ten years. (Chang, Minwoo, Marc Maguire, and Yan Sun,2019).

Artificial Neural Network is another tool that is widely used for predictive analysis. However, the linear regression model against the ANN, the ANN being non-linear, resembles the bridge deterioration more appropriately. Due to this, the accuracy of the neural network is higher than the linear regression model.

Analyzing the different attributes in the datasheet, we know that a classification of various components can be developed for further study. For instance, the first explanatory variable is the wearing surface for the deck elements, which includes ten types. The second is the span length of the bridge, classified into five categories for a span size every 50 meters. The following variables are the functional classification of inventory route, average daily traffic, annual average daily traffic, and the number of lanes. We know the distribution of these bridges into different categories based on

other explanatory variables. (Chang, Minwoo, Marc Maguire, and Yan Sun,2019). The Average Daily Traffic plays an essential role in modeling bridge elements.

Algorithms used in this paper to determine the significant features influencing the target variable.

Random Forest Classifier

Ensemble learning is a type of machine learning method where combination of various weak learners is done. Random Forest is a supervised machine learning algorithm formed by the ensemble of decision trees. The random forest classifier provides improved results by combining all the decision trees and providing flexible results. It is usually trained with a bagging method which improves the prediction result based on the combination of learning models.(Datacamp. 2022. "Sklearn random forest classifiers in Python tutorial." Accessed April 12, 2021. https://www.datacamp.com/community /tutorials/random-forests-classifier-python/). Being versatile and straightforward, random forest algorithm produces effective prediction results (Rajkumar, Monica, Sudhagar Nagarajan, and Madasamy Arockiasamy, 2023). Although this algorithm does not overfit, it is a time - consuming process since various decision trees are involved in making the decision. It works on the basis of five basic steps:

1. From the input dataset, random samples get selected.
2. For every sample a decision tree is constructed.
3. Prediction result is fetched from each decision tree.
4. Voting is performed for each predicted result.

5. The prediction result with highest votes is selected as final prediction.

(Rajkumar, Monica, Sudhagar Nagarajan, and Madasamy Arockiasamy, 2023).

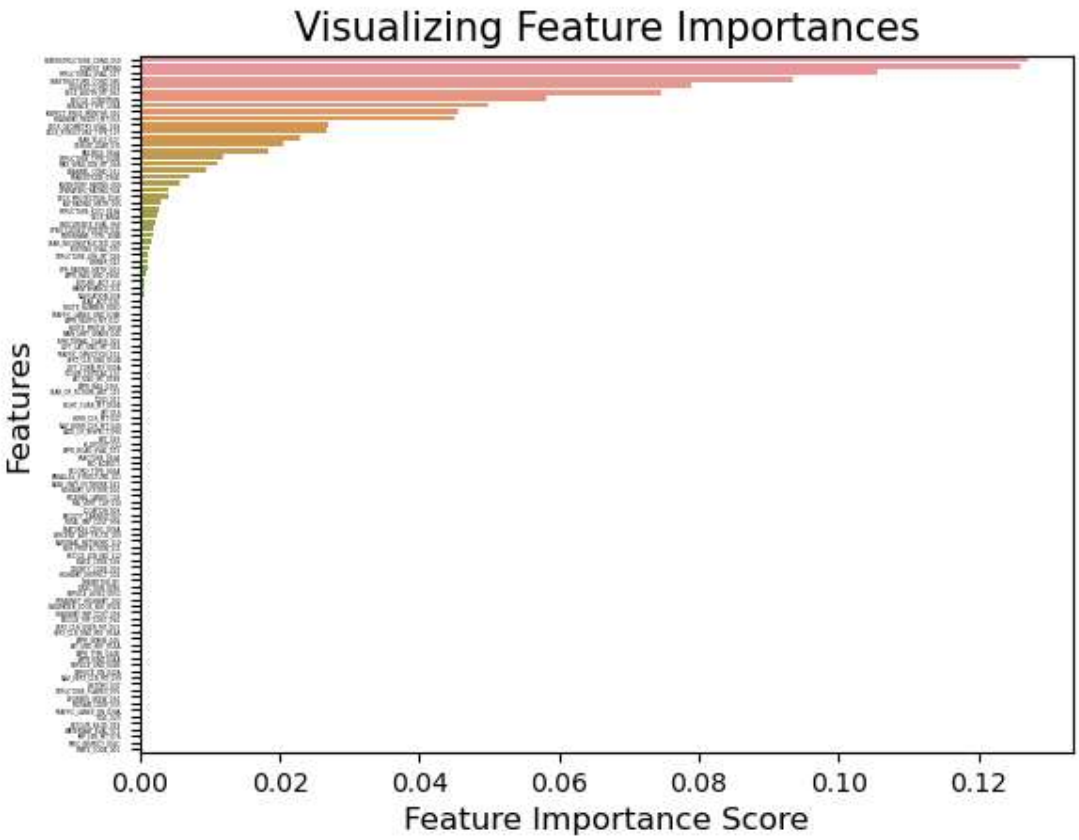


Figure 3 : Visualizing the Feature Importances of 108 parameters on Deck Condition of the bridge using Random Forest algorithm

Table 5: Top 25 parameters using Random Forest (Gini Coefficient) using sklearn library,

1	SUPERSTRUCTURE_COND_059	0.127199
2	LOWEST_RATING	0.126154
3	STRUCTURAL_EVAL_067	0.105467
4	SUBSTRUCTURE_COND_060	0.093557

5	CULVERT_COND_062	0.078992
6	DECK_WIDTH_MT_052	0.074599
7	BRIDGE_CONDITION	0.058059
8	SURFACE_TYPE_108A	0.049827
9	INSPECT_FREQ_MONTHS_091	0.045536
10	ROADWAY_WIDTH_MT_051	0.045062
11	DECK_GEOMETRY_EVAL_068	0.027065
12	DECK_STRUCTURE_TYPE_107	0.026694
13	YEAR_BUILT_027	0.023004
14	DESIGN_LOAD_031	0.020375
15	RAILINGS_036A	0.018242
16	STRUCTURE_TYPE_043B	0.011884
17	MAX_SPAN_LEN_MT_048	0.011031
18	CHANNEL_COND_061	0.009556
19	TRANSITIONS_036B	0.006989
20	INVENTORY_RATING_066	0.005774
21	OPERATING_RATING_064	0.004108
22	DECK_PROTECTION_108C	0.004036
23	INV_RATING_METH_065	0.002997
24	STRUCTURE_KIND_043A	0.002804
25	DECK_AREA	0.002395

Table 6 : Top 5 parameters using Random Forest (Gini Coefficient) using sklearn library

1	SUPERSTRUCTURE_COND_059	0.127199
2	LOWEST_RATING	0.126154

3	STRUCTURAL_EVAL_067	0.105467
4	SUBSTRUCTURE_COND_060	0.093557
5	CULVERT_COND_062	0.078992

Table 7: Accuracy of both models using Random Forest

Model	Input Parameters	Accuracy
Random_Forest_25_Params	25	0.7626
Random_Forest_5_Params	5	0.7917

Generalized Linear Regression

A generalized linear regression model is made of a linear predictor

$$\eta_i = \beta_0 + \beta_1\chi_1 + \ldots + \beta_p x_{pi}$$

consisting of a link function which describes how the mean, E (Y_i) = μ_i, depends on the linear predictor,

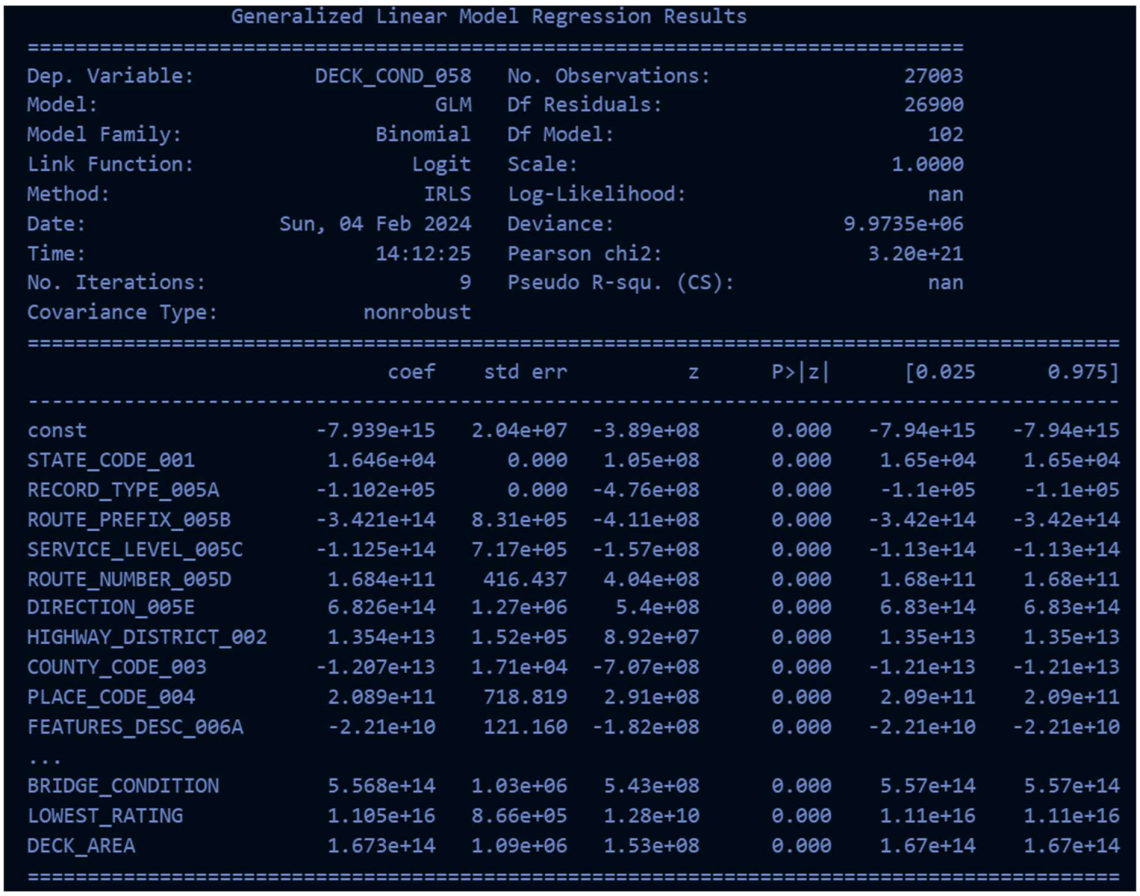
$$g(\mu_i) = \eta_i$$

and variance function that depends on how the variance $Var(Y_i)$ depends on the mean,

$$Var(Y_i) = \phi V(\mu)$$

where the dispersion parameter ϕ is a constant. (Turner, Heather, 2008)

To fit a regression between the target variable 'Deck_condition_059' and input variables, the Statsmodels library has been used to know the influencing parameters.



Generalized Linear Model Regression Results

Dep. Variable:	DECK_COND_058	No. Observations:	27003
Model:	GLM	Df Residuals:	26900
Model Family:	Binomial	Df Model:	102
Link Function:	Logit	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	nan
Date:	Sun, 04 Feb 2024	Deviance:	9.9735e+06
Time:	14:12:25	Pearson chi2:	3.20e+21
No. Iterations:	9	Pseudo R-squ. (CS):	nan
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	-7.939e+15	2.04e+07	-3.89e+08	0.000	-7.94e+15	-7.94e+15
STATE_CODE_001	1.646e+04	0.000	1.05e+08	0.000	1.65e+04	1.65e+04
RECORD_TYPE_005A	-1.102e+05	0.000	-4.76e+08	0.000	-1.1e+05	-1.1e+05
ROUTE_PREFIX_005B	-3.421e+14	8.31e+05	-4.11e+08	0.000	-3.42e+14	-3.42e+14
SERVICE_LEVEL_005C	-1.125e+14	7.17e+05	-1.57e+08	0.000	-1.13e+14	-1.13e+14
ROUTE_NUMBER_005D	1.684e+11	416.437	4.04e+08	0.000	1.68e+11	1.68e+11
DIRECTION_005E	6.826e+14	1.27e+06	5.4e+08	0.000	6.83e+14	6.83e+14
HIGHWAY_DISTRICT_002	1.354e+13	1.52e+05	8.92e+07	0.000	1.35e+13	1.35e+13
COUNTY_CODE_003	-1.207e+13	1.71e+04	-7.07e+08	0.000	-1.21e+13	-1.21e+13
PLACE_CODE_004	2.089e+11	718.819	2.91e+08	0.000	2.09e+11	2.09e+11
FEATURES_DESC_006A	-2.21e+10	121.160	-1.82e+08	0.000	-2.21e+10	-2.21e+10
...						
BRIDGE_CONDITION	5.568e+14	1.03e+06	5.43e+08	0.000	5.57e+14	5.57e+14
LOWEST_RATING	1.105e+16	8.66e+05	1.28e+10	0.000	1.11e+16	1.11e+16
DECK_AREA	1.673e+14	1.09e+06	1.53e+08	0.000	1.67e+14	1.67e+14

Figure 4: Parameters influencing target variable using Generalized Linear Model Regression

Table 8: Top 25 parameters in descending order influencing the deck condition according to Generalized Linear Regression are:

1	STATE_CODE_001
2	RECORD_TYPE_005A
3	ROUTE_PREFIX_005B
4	SERVICE_LEVEL_005C
5	ROUTE_NUMBER_005D
6	DIRECTION_005E
7	HIGHWAY_DISTRICT_002
8	COUNTY_CODE_003

9	PLACE_CODE_004
10	FEATURES_DESC_006A
11	FACILITY_CARRIED_007
12	LOCATION_009
13	MIN_VERT_CLR_010
14	KILOPOINT_011
15	BASE_HWY_NETWORK_012
16	LAT_016
17	LONG_017
18	DETOUR_KILOS_019
19	TOLL_020
20	MAINTENANCE_021
21	OWNER_022
22	FUNCTIONAL_CLASS_026
23	YEAR_BUILT_027
24	TRAFFIC_LANES_ON_028A
25	TRAFFIC_LANES_UND_028B

Table 9: Top 5 parameters influencing deck condition in descending order according to Generalized Linear Regression

1	STATE_CODE_001
2	RECORD_TYPE_005A
3	ROUTE_PREFIX_005B
4	SERVICE_LEVEL_005C
5	ROUTE_NUMBER_005D

Table 10: Accuracy of the model using Generalized Linear Regresssion

Model	Input Parameters	Accuracy
Ordinary_Least_Squares_25_Params	25	0.3238
Ordinary_Least_Squares_5_Params	5	0.3228

Ordinal Logistic Regression

OLS Regression Results

Dep. Variable:	DECK_COND_058	R-squared (uncentered):	0.986
Model:	OLS	Adj. R-squared (uncentered):	0.986
Method:	Least Squares	F-statistic:	1.875e+04
Date:	Sun, 04 Feb 2024	Prob (F-statistic):	0.00
Time:	14:37:43	Log-Likelihood:	-29233.
No. Observations:	27003	AIC:	5.867e+04
Df Residuals:	26901	BIC:	5.951e+04
Df Model:	102		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
STATE_CODE_001	-5.22e-14	3.76e-14	-1.387	0.166	-1.26e-13	2.16e-14
RECORD_TYPE_005A	-1.213e-13	6.05e-14	-2.007	0.045	-2.4e-13	-2.81e-15
ROUTE_PREFIX_005B	-0.0161	0.009	-1.821	0.069	-0.034	0.001
SERVICE_LEVEL_005C	-0.0036	0.008	-0.470	0.638	-0.018	0.011
ROUTE_NUMBER_005D	1.048e-05	4.43e-06	2.366	0.018	1.8e-06	1.92e-05
DIRECTION_005E	0.0243	0.013	1.812	0.070	-0.002	0.050
HIGHWAY_DISTRICT_002	0.0031	0.002	1.917	0.055	-6.98e-05	0.006
COUNTY_CODE_003	-0.0001	0.000	-0.670	0.503	-0.000	0.000
PLACE_CODE_004	6.01e-06	7.66e-06	0.784	0.433	-9.01e-06	2.1e-05
FEATURES_DESC_006A	-5.002e-07	1.29e-06	-0.387	0.698	-3.03e-06	2.03e-06
FACILITY_CARRIED_007	-7.897e-07	1.22e-06	-0.648	0.517	-3.18e-06	1.6e-06

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[1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[3] The smallest eigenvalue is 8e-20. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Figure 6: Parameters influencing taret variable using OLS regression

Table 11: Top 25 parameters in descending order influencing deck condition according to ordinal logistic regression

1	MEDIAN_CODE_033
2	RECORD_TYPE_005A
3	DECK_WIDTH_MT_052
4	DESIGN_LOAD_031
5	SPEC_INSPECT_092C
6	DETOUR_KILOS_019
7	ROUTE_NUMBER_005D
8	UNDCLRENCE_EVAL_069
9	APPR_RAIL_END_036D
10	SERVICE_ON_042A
11	NATIONAL_NETWORK_110
12	LEFT_LAT_UND_MT_056
13	SCOUR_CRITICAL_113
14	DATE_OF_INSPECT_090
15	NAV_VERT_CLR_MT_039
16	APPR_RAIL_036C
17	APPR_KIND_044A
18	BRIDGE_CONDITION
19	MAX_SPAN_LEN_MT_048
20	APPR_TYPE_044B
21	TRAFFIC_DIRECTION_102
22	YEAR_BUILT_027
23	CHANNEL_COND_061

24	MEMBRANE_TYPE_108B
25	INV_RATING_METH_065

Table 9: Top 5 parameters in descending order influencing deck condition

1	STATE_CODE_001
2	RECORD_TYPE_005A
3	ROUTE_PREFIX_005B
4	SERVICE_LEVEL_005C
5	ROUTE_NUMBER_005D

Table 12: Accuracy of models based on ordinal logistic model

Model	Input Parameters	Accuracy
Ordinal_Logistic_Regression_25_Params	25	0.4996
Ordinal_Logistic_Regression_5_Params	5	0.3266

Principal Component Analysis

Table 13: Top 5 parameters that significantly influence target variable are

SUPERSTRUCTURE_COND_059	0.127199
LOWEST_RATING	0.126154
STRUCTURAL_EVAL_067	0.105467
SUBSTRUCTURE_COND_060	0.093557

CULVERT_COND_062	0.078992

Table 14: Accuracies of both the models

Model	Input Parameters	Accuracy
Principal_Componant_Analysis_25_Params	25	0.3265
Principal_Componant_Analysis_5_Params	5	0.3235

Negative Binomial Method

Generalized Linear Model Regression Results

Dep. Variable:	DECK_COND_058	No. Observations:	27003
Model:	GLM	Df Residuals:	26901
Model Family:	NegativeBinomial	Df Model:	101
Link Function:	Log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-72647.
Date:	Sun, 04 Feb 2024	Deviance:	1486.4
Time:	16:42:23	Pearson chi2:	979.
No. Iterations:	100	Pseudo R-squ. (CS):	0.2405
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
STATE_CODE_001	-1.38e-14	5.63e-14	-0.245	0.807	-1.24e-13	9.66e-14
RECORD_TYPE_005A	-1.438e-14	1.8e-13	-0.080	0.936	-3.68e-13	3.39e-13
ROUTE_PREFIX_005B	-0.0044	0.014	-0.308	0.758	-0.032	0.024
SERVICE_LEVEL_005C	-0.0237	0.012	-2.006	0.045	-0.047	-0.001
ROUTE_NUMBER_005D	1.027e-05	6.95e-06	1.478	0.139	-3.34e-06	2.39e-05
DIRECTION_005E	-0.0255	0.021	-1.234	0.217	-0.066	0.015
HIGHWAY_DISTRICT_002	-0.0013	0.003	-0.529	0.597	-0.006	0.004
COUNTY_CODE_003	-0.0003	0.000	-1.069	0.285	-0.001	0.000
PLACE_CODE_004	-3.301e-06	1.2e-05	-0.276	0.783	-2.68e-05	2.02e-05
FEATURES_DESC_006A	-1.419e-06	2.03e-06	-0.699	0.484	-5.4e-06	2.56e-06
FACILITY_CARRIED_007	-1.817e-06	1.9e-06	-0.955	0.340	-5.55e-06	1.91e-06
...						
BRIDGE_CONDITION	-0.0290	0.017	-1.668	0.095	-0.063	0.005
LOWEST_RATING	-0.0532	0.015	-3.485	0.000	-0.083	-0.023
DECK_AREA	-0.0105	0.018	-0.591	0.554	-0.046	0.024

Figure 7: Parameters influencing target variable using Negative Binomial algorithm

There are 23 parameters that have been chosen instead of 25 for Negative Binomial since there are only 23 parameters that are above the 95% confidence interval influencing the target variable.

Table 15: Top 23 parameters and not 25 influencing deck condition using the Negative Binomial since only 23 parameters exist, which are significant as the p–value is less than 0.05 or a confidence interval of 95%.

CULVERT_COND_062	4.14E-96
SUPERSTRUCTURE_COND_059	1.91E-91
SUBSTRUCTURE_COND_060	6.57E-27
TOLL_020	7.28E-15
TRAFFIC_DIRECTION_102	3.15E-13
TRAFFIC_LANES_ON_028A	2.88E-12
STRUCTURAL_EVAL_067	7.32E-12
DECK_GEOMETRY_EVAL_068	1.33E-11
ROADWAY_WIDTH_MT_051	3.91E-11
VERT_CLR_OVER_MT_053	1.80E-10
MIN_VERT_CLR_010	6.54E-07
DECK_STRUCTURE_TYPE_107	8.96E-05
LOWEST_RATING	4.39E-04
SURFACE_TYPE_108A	1.82E-03
APPR_WIDTH_MT_032	5.57E-03
UNDCLRENCE_EVAL_069	9.19E-03
RAILINGS_036A	1.07E-02
HISTORY_037	1.11E-02
OPEN_CLOSED_POSTED_041	1.79E-02

POSTING_EVAL_070	2.33E-02
HORR_CLR_MT_047	2.90E-02
SERVICE_LEVEL_005C	4.20E-02
FRACTURE_092A	4.57E-02

Table 14: Top 5 parameters influencing the deck condition of the bridge using the Negative Binomial algorithm are:

CULVERT_COND_062	4.14E-96
SUPERSTRUCTURE_COND_059	1.91E-91
SUBSTRUCTURE_COND_060	6.57E-27
TOLL_020	7.28E-15
TRAFFIC_DIRECTION_102	3.15E-13

Table 15: Accuracies of models using both 25 and 5 parameters

Model	Input Parameters	Accuracy
Negative_Binomial_25_Params	23	0.8052
Negative_Binomial_5_Params	5	0.7144

Building a customized model for accuracy improvement beyond 0.8052, which is the highest as observed after determining the top 25 significant parameters using Negative Binomial and base neural network. Keras Tuner can be used for hyperparameter tuning which is an in-built model.

Conclusion

1. Comparison of different algorithms in combination with neural networks to identify the best most influencing parameters has been exhaustively dealt in this paper.
2. Algorithms such as Random Forest Model, Generalized Linear Model, Ordinal Logistic Regression and Principal Component Analysis have been used to identify the top 25 and 5 significant features separately influencing the deck condition.
3. Negative Binomial was selected for the reason of showing a higher accuracy, which in combination with neural network was used as final model for prediction of deck condition after
4. Hyperparameter optimization using keras tuner to improve the accuracy of the model
5. Accuracy of 80% was reached by the model using negative binomial and artificial neural network to predict the deck condition.
6. Some algorithms such as Random Forest have shown a higher accuracy with 5 parameters as input variables in comparison to 25 parameters as input which is contrary to general perception of higher input variables showcasing a higher accuracy.

References

1. Agrawal, A. K., Kawaguchi, A., & Chen, Z. (2010). Deterioration rates of typical bridge elements in New York. *Journal of Bridge Engineering*, *15*(4), 419-429.
2. Ali, G., Elsayegh, A., Assaad, R., El-adaway, I. H., & Abotaleb, I. S. (2019). Artificial neural network model for bridge deterioration and asse
3. Althaqafi, E., & Chou, E. (2022). Developing Bridge Deterioration Models Using an Artificial Neural Network. *Infrastructures*, *7*(8), 101.
4. Analysis of National Bridge Inventory (NBI) Data for California Bridges By Emily Yu UC Connect Summer Research 2015
5. Assaad, R., & El-adaway, I. H. (2020). Evaluation and prediction of the hazard potential level of dam infrastructures using computational artificial intelligence algorithms. *Journal of Management in Engineering*, *36*(5), 04020051.
6. Ayyub, B. M., & McCuen, R. H. (2003). Probability, statistics, and reliability for engineers and scientists. *Technometrics*, *45*(3), 276.
7. Chang, M., Maguire, M., & Sun, Y. (2019). Stochastic modeling of bridge deterioration using classification tree and logistic regression. *Journal of Infrastructure Systems*, *25*(1), 04018041.
8. Chugh, G., Kumar, S., & Singh, N. (2021). Survey on machine learning and deep learning applications in breast cancer diagnosis. *Cognitive Computation*, *13*(6), 1451-1470.
9. Contreras-Nieto, C., Shan, Y., & Lewis, P. (2018). Characterization of steel bridge superstructure deterioration through data mining techniques. *Journal of Performance of Constructed Facilities*, *32*(5), 04018062

10. Datacamp. 2022. “Sklearn random forest classifiers in Python tuto rial.” Accessed April 12, 2021. https://www.datacamp.com/community /tutorials/random-forests-classifier-python/ .
11. Dawood, T., Z. Zhu, and T. Zayed. 2018. “Computer vision–based model for moisture marks detection and recognition in subway networks.” J. Comput. Civ. Eng. 32 (2): 04017079. https://doi.org/10.1061/(ASCE)CP .1943-5487.0000728
12. Dhada, M., Hadjidemetriou, G. M., & Parlikad, A. K. (2020). Predicting bridge elements deterioration, using collaborative gaussian process regression. *IFAC-PapersOnLine*, *53*(3), 348-353.
13. Huang, Y. H. (2010). Artificial neural network model of bridge deterioration. *Journal of Performance of Constructed Facilities*, *24*(6), 597-602.
14. Khuriwal, N., & Mishra, N. (2018, October). Breast cancer diagnosis using deep learning algorithm. In *2018 International Conference on Advances in Computing, Communication Control and Networking (ICACCCN)* (pp. 98-103). IEEE
15. La, H. M., Gucunski, N., Dana, K., & Kee, S. H. (2017). Development of an autonomous bridge deck inspection robotic system. *Journal of Field.*
16. Li, Q.-F.; Song, Z.-M. High-performance concrete strength prediction based on ensemble learning. Constr. Build. Mater. 2022, 324, 126694. [CrossRef]
17. Mahmoud, O., Anwar, F., & Salami, M. J. E. (2007). Learning algorithm effect on multilayer feed forward artificial neural network performance in image coding.
18. Negnevitsky, M. (2005). *Artificial intelligence: a guide to intelligent systems*. Pearson education.
19. Nelder, J. A., & Wedderburn, R. W. (1972). Generalized linear models. *Journal of the Royal Statistical Society Series A: Statistics in Society*, *135*(3), 370-384.
20. Prasad, A., Juran, I., & Yanev, B. (2022). Statistical data analysis of the aging effect on time-dependent corrosion rate for steel girder bridges with concrete decks. *Journal of Infrastructure Systems*, *28*(2), 04022002.
21. Rajkumar, M., Nagarajan, S., & Arockiasamy, M. (2023). Bridge Infrastructure Management System: Autoencoder Approach for Predicting Bridge Condition Ratings. *Journal of Infrastructure Systems*, *29*(1), 04022042

22. Solorio-Fernández, S., Carrasco-Ochoa, J. A., & Martínez-Trinidad, J. F. (2020). A review of unsupervised feature selection methods. *Artificial Intelligence Review*, *53*(2), 907-948.
23. Srinivasan, D., Chang, C. S., & Liew, A. C. (1995). Demand forecasting using fuzzy neural computation, with special emphasis on weekend and public holiday forecasting. *IEEE Transactions on Power Systems*, *10*(4), 1897-1903.
24. Turner, H. (2008). Introduction to generalized linear models. *Rapport technique, Vienna University of Economics and Business.*
25. Vincent, D. R., N. Deepa, D. Elavarasan, K. Srinivasan, S. H. Chauhdary, and C. Iwendi. 2019. "Sensors driven AI-based agriculture recommen dation model for assessing land suitability." Sensors 19 (17): 3667. https://doi.org/10.3390/s19173667